MIL-D-82134B(MC)
12 June 1973
SUPERSEDING
MIL-D-82134A(MC)
3 March 1970

MILITARY SPECIFICATION

DUMMY LOADS, ELECTRICAL: 60/100 KW, 55/90 KVAR, 3 PHASE, 60/400 HERTZ, AC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for transportable dummy loads designed to simulate the combined electrical characteristics of a generator set load and to dissipate the energy of the load circuit.

1.2 Classification. The dummy loads shall be of the following types, as specified (see 6.2).

Type I - 60 kilowatt (kW), 55 kilovar (kvar), 3 phase, 60/400 hertz (Hz), alternating current (ac).

Type II - 100 kW, 90 kvar, 3 phase, 60/400 Hz, ac.

Unless otherwise specified herein, requirements shall apply to both types.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of the invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-485 - Enamel, Semi-gloss, Rust-inhibiting.

FSC 6115

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MILITARY

MIL-T-704	-	Treatment And Painting Of Materiel.
MIL-E-16298	-	Electric Machines Having Rotating Parts And Associated Repair Parts: Packaging Of.
MIL-E-16400	-	Electronic Equipment, Naval Ship And Shore, General Specification.
MIL-C-45150	-	Chassis, Trailer 2-Wheel Cart Type 1/4 to 3-1/2 Ton.

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Characteristics Requirement for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	-	Environmental Test Methods.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS
Codes of Electrical and Electronic Engineers

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, New York 10007.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS
Codes of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 29 West 29th Street, New York, New York 10018.)

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AMERICAN INSTITUTE OF STEEL CONSTRUCTION
Codes of the American Institute of Steel Construction

(Application for copies should be addressed to the American Institute of Steel Construction, 25 West 39th Street, New York, New York 10018.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION
Codes of National Electrical Manufacturers Association

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, New York 10017.)

NATIONAL BOARD OF FIRE UNDERWRITERS
Codes of National Board for Fire Underwriters

(Application for copies should be addressed to the National Board of Fire Underwriters, 85 John Street, New York, New York 10038.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, 1616 "P" Street, N. W. Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), prior to submission of any complete production items for acceptance, the contractor shall furnish one first article for determination of conformance to this specification. Examination and tests shall be those specified herein. Approval of the first article by the procuring activity shall not relieve the contractor of his obligation to supply dummy loads conforming to this specification. Any changes or deviations of the production

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items from the first article shall be subject to the approval of the contracting officer. The first article may be included in the quantity to be delivered to the Government providing it is refurbished and otherwise meets the requirements specified herein.

3.2 Materials and parts. Unless otherwise specified herein, all materials and parts used in the construction of the dummy load shall be in accordance with MIL-E-16400.

3.2.1 Fungus-inert materials. Requirement 4 of MIL-STD-454 applies for fungus-inert materials used in the dummy load.

3.2.2 Unacceptable materials. Materials specified as unacceptable in MIL-E-16400 shall not be used in construction of the dummy load.

*3.3 Design. The dummy loads shall be designed as skid mounted transportable units which provide a reactive load of 55 kvar and a resistive load of 60 kW for the type I dummy load and a reactive load of 90 kvar and a resistive load of 100 kW for the type II dummy load. The dummy loads shall provide provisions for 3-phase, 4-wire, 60- and 400-Hz 120/208- and 240/416-volt (V) power input. A control panel for selection of various combinations of resistive and reactive loads shall be provided. The dummy load shall be capable of operating with 25 percent overvoltage applied for intermittent periods of up to 30 seconds.

3.3.1 Safety provisions. Provisions for safety of personnel shall be provided in accordance with requirement 1 of MIL-STD-454. Operation in damp and rainy weather shall be possible without hazard to operating personnel.

3.3.2 Enclosure. The dummy load shall be enclosed in a sheet steel enclosure of 18-gage minimum thickness reinforced by an internal steel framework. Design of the enclosure shall be adequate to withstand requirements specified herein. The enclosure shall be designed to prevent water, sand and dust which would impair the proper operation of the dummy load from collecting within the enclosure.

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3.3.2.1 Covers and doors. Weathertight covers and doors shall be provided for all openings required for operation and maintenance of the dummy load. Covers and doors which are required to be open during operation shall be hinged and shall be equipped with positive retaining devices to hold them in the open position. Covers and doors required to be opened only for maintenance shall be completely removable rather than hinged. No covers or doors shall be required on the top surface of the dummy load enclosure. The number of separate covers and doors required shall be minimized. Fastening devices shall be in accordance with MIL-E-16400. Fastening devices on covers and doors which are required to be open for operation shall be of a type which is captive to the cover or door.

*3.3.2.2 Storage compartment. A storage compartment shall be provided for storage of auxiliary power cables and instruction manuals. The dimensions of the compartment shall be 12 ±2 inches high by 28 ±2 inches wide by 15 ±2 inches deep. The compartment shall be equipped with convenient doors and fasteners.

3.3.2.3 Base. The dummy load shall be equipped with a base which shall be an integral structural part of the enclosure. The bottom of the base shall have a contour surface.

3.3.2.4 Skids. Two replaceable skids shall be provided. Each skid shall be 3 inches high by 3 inches wide and shall extend the full length of the dummy load but not beyond the ends of the enclosure. The skids shall be attached to the base in a manner which will allow replacement with common hand tools from the outside of the dummy load. Towing eyes shall be provided on both ends of the skids. Each towing eye shall have an inside diameter of not less than 2 inches. This may be accomplished with use of a clevis (a minimum of two to be supplied). The towing eyes shall be capable of withstanding a 7,000-pound tensile load when applied in any direction. Slots shall be provided to allow forklifting of the dummy load from either side.

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3.3.3 Drainage provisions. Drainage provisions shall be provided to prevent water which would impair proper operation from collecting within the dummy load.

*3.3.4 Lifting and tiedown. Combination lifting and tiedown eyes shall be provided on each of the four upper corners of the dummy load. Design of the eye assemblies shall be such that lifting and tiedown cables will not damage the dummy load during lifting or tiedown for transport. The lifting and tiedown eyes shall be capable of withstanding a minimum 7,000-pound tensile load when applied in any direction.

3.3.5 Trailer mounting provisions. The dummy load shall be designed to mount on a M-353, 3-1/2-ton trailer conforming to MIL-C-45150.

3.3.6 Electrical grounding. An easily accessible ground lug shall be provided on the dummy load. A 6-foot ground cable and a 6-foot "T" handle copper or aluminum grounding rod shall also be provided. Stowage provisions shall be provided within or on the outside of the enclosure for the grounding rod. The grounding rod shall be a single section with removable handle. Appropriate cable clamps shall be provided.

3.3.7 Control panel. A control panel shall be provided. The layout and controls to be provided for type I dummy load shall be in accordance with Figure 1. Additional controls shall be provided to meet requirements specified herein for type II dummy load. The controls provided shall allow infinite adjustment across the entire rated load range (both reactive and resistive) of the dummy load. The controls shall be grouped by phases as depicted in Figure 1. The load element switches shall be toggle type.

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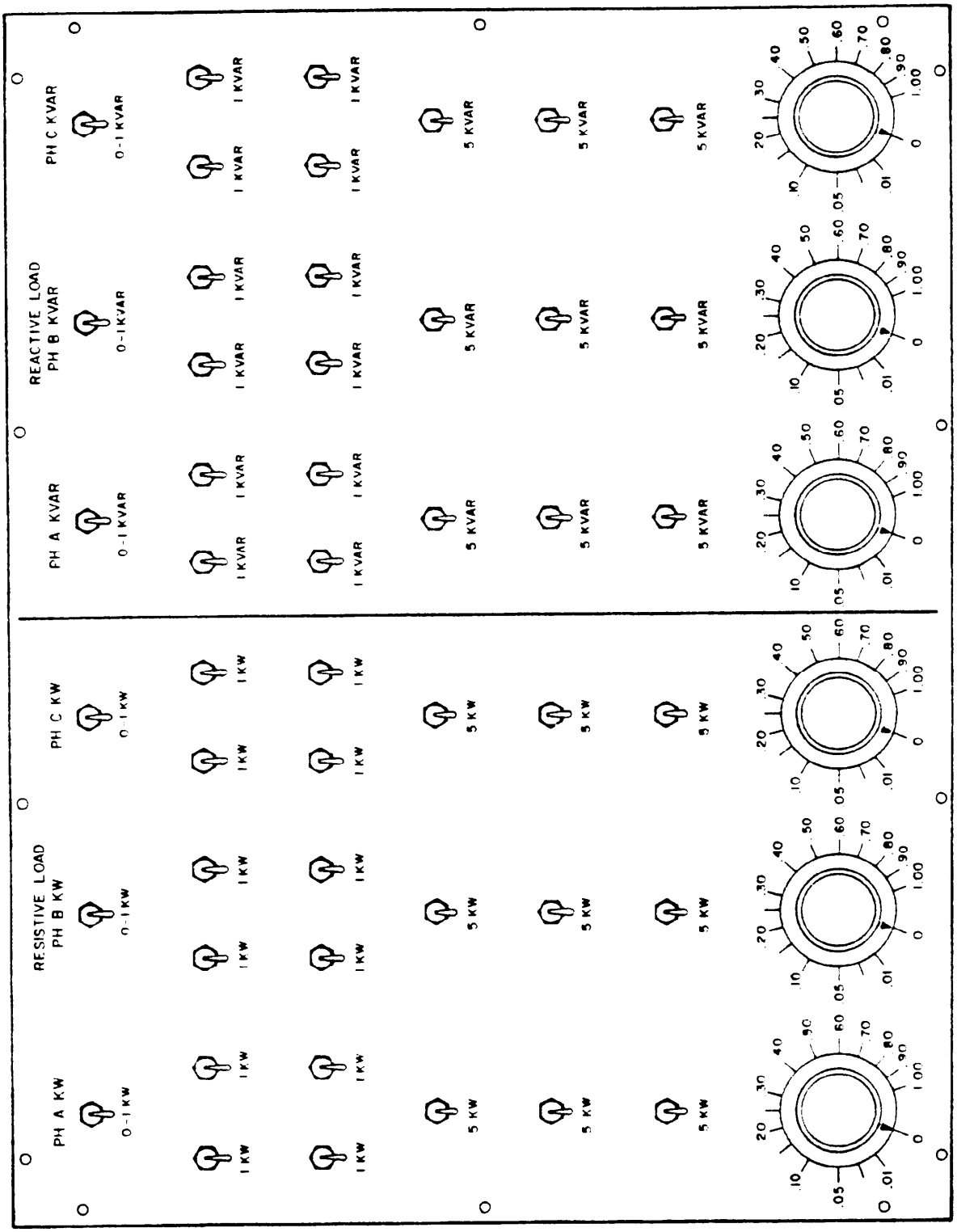


Figure 1 -- Control Panel

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*3.3.8 Metering and indicator panel. A metering and indicator panel shall be provided and shall be mounted on the dummy load directly above the control panel. Meters, switches, indicators and test jacks shall be provided and the panel layout shall be in accordance with Figure 2 except, ammeters, kilovoltampere (kVA) meters and associated range selector switches shall have a minimum of four ranges each. A phase sequence indicator shall also be provided and shall be connected directly to the power input terminals to indicate whether or not the power being applied is of the proper phase sequence. The indicator shall operate from all voltage and frequency inputs (see 3.3). The indicator shall be installed on the panel near the frequency meter selector switch. Each meter shall have an accuracy of 2 percent of full scale or better. Test jacks for calibration of each meter shall be provided. Protective circuitry shall be provided for all meters to prevent damage to the meters for overvoltage inputs of up to 200 percent. The power factor (pf) meter shall cover the minimum range of 0.3 to 1. The voltmeters shall cover the voltage ranges specified in 3.3. The ammeters and kVA meter shall cover the appropriate ranges for the type dummy load ordered (see 1.2 and 6.2).

3.3.9 Resistive load. Resistive load elements shall be furnished as required to provide the kW load specified for the type dummy load ordered (see 1.2 and 6.2). The number of resistive elements provided and the circuit configuration shall be consistent with requirements of 3.3.8. The cold to hot resistance shall not vary more than 3 percent.

*3.3.10 Reactive load. Inductors shall be furnished as required to provide kvar load specified for the type dummy load ordered (see 1.2 and 6.2). The number of inductors provided and the circuit configuration shall be consistent with requirements of 3.3.7. The inductors shall be of an iron core construction with an air gap which limits the flux density in the iron to a value well below the saturation point. Wave form distortion shall not exceed one-half of one percent. The pf of the inductors shall not exceed 0.05 at 400 Hz.

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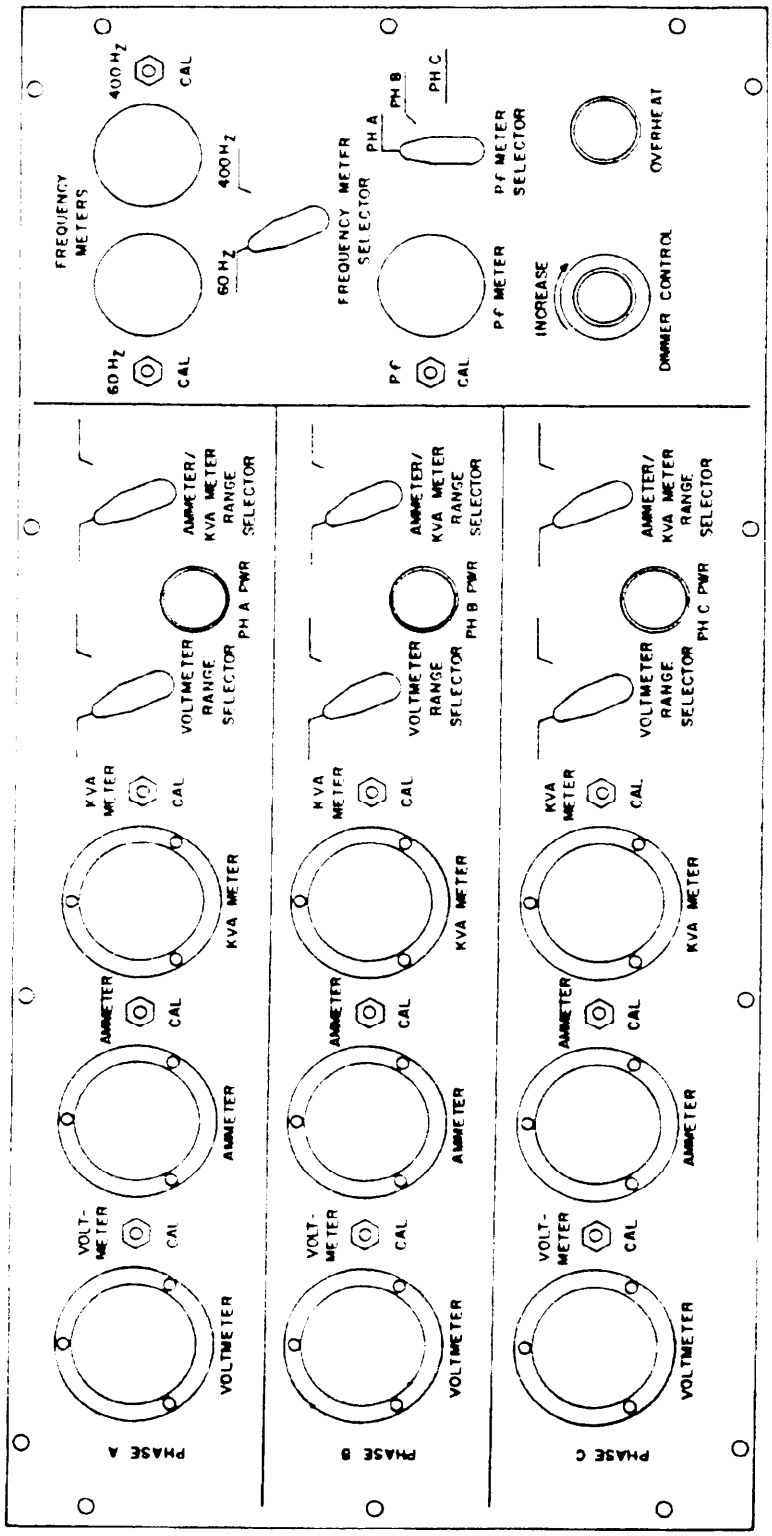


Figure 2 — Metering and Indicator Panel

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*3.3.11 Panel illumination. The panels (see 3.3.7 and 3.3.8) shall be adequately illuminated to permit operation when in total darkness. Lights shall be manually operated by a single switch. An indicator light shall be provided for each phase to indicate when power is being applied. All lights shall be dimmer controlled, either optically or electrically, from zero to full brilliance.

3.3.12 Internal cooling provisions. An electric motor driven fan shall be provided for internal cooling. The motor shall be capable of operating from all input power (both frequency and voltage) specified in 3.3. In addition, the motor shall be capable of operation from an external 120/208-V, 3-phase, 60-Hz commercial power source. Appropriate external power input provisions and selector switch shall be provided. An air pressure sensing device shall be provided and connected in a manner which will automatically disconnect dummy load input power in the event the cooling fan fails to operate or if the cooling air path is obstructed.

3.3.13 Power input provisions. Power input controls and terminals shall be located on the right side (when facing control panel) and toward the control panel end of the dummy load.

*3.3.14 Load terminals. A terminal board with four studs and nuts shall be provided for connection of lug-terminated cables. The terminal studs and nuts shall be made of corrosion-resistant material and shall be of sufficient size to carry twice the maximum rated current when the dummy load is connected for 120/208-V operation. The studs shall be of sufficient length to accommodate two lug-terminated cables to permit parallel operation of two generator sets. The terminal studs shall be constructed and mounted in the terminal board so they will not turn when the terminal stud nuts are tightened or loosened. The terminal studs shall be marked L_1, L_2, L_3 and L_0.

3.3.15 Power input controls. The total power input (both reactive and resistive) shall be controlled and protected by a 3-phase circuit breaker or contactor of the appropriate rating. The rating of the circuit breaker or contactor shall be predicated on the type dummy load ordered (see 1.2 and 6.2).

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3.3.16 Voltage input selector. A selector shall be provided for 120/208- or 240/416-V operation. The selector shall be a plug-in type or a single lever switch type. Protective provisions shall be provided to protect the dummy load in the event a 240/416-V input is applied when the voltage input selector is in the 120/208-V position.

*3.3.17 Frequency selector. A frequency selector for selection of 60- or 400-Hz operation shall be provided. If more than one selector switch is used, mechanical interconnection between the switches shall be provided. Protective circuitry shall be provided to protect the dummy load in the event the wrong frequency is applied.

3.3.18 Interchangeability. Interchangeability shall be in accordance with requirement 7 of MIL-STD-454.

3.4 Construction. The dummy load shall be constructed in accordance with the following codes and shall be adequate to meet all requirements specified herein.

American Society of Mechanical Engineers

Institute of Electrical and Electronic Engineers

American Institute of Steel Construction

National Electrical Manufacturers Association

National Board of the Fire Underwriters.

3.4.1 Dripproof construction. The dummy load, including control and meter panels, shall be dripproof in accordance with MIL-E-16400. All electrical components shall be dripproof as specified above or shall be of a type not affected by moisture.

3.5 Maintainability. Adjustment, repair and replacement of parts, components, and assemblies shall be possible with minimum disassembly and removal of other parts and assemblies. All repair shall be possible with use of common hand tools.

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3.6 Transportability. The dummy load shall be capable of withstanding loads imposed when transported by railroad flat car, by military vehicles over rough terrain and by air transport, including helicopter lift.

*3.7 Performance. The dummy load shall be capable of proper operation specified herein under the following conditions or after having been subjected to such conditions:

(a) Temperature. -40°C to 55°C operating and -60°C to 71°C nonoperating.

(b) Humidity. 0 to 95 percent relative over entire operating temperature range.

(c) Salt fog. In accordance with MIL-STD-810, method 509.

(d) Rain. In accordance with MIL-STD-810, method 506 during operation. In accordance with MIL-E-16400 for dripproof enclosures, nonoperating.

(e) Vibration. Vibration as encountered when the dummy load is mounted on an M-353 trailer (see 3.6) and towed over rough terrain for distances up to 500 miles.

(f) Shock. Drop from a height of 18 inches onto a concrete surface.

3.8 Radio interference suppression. The dummy load shall meet the radiated radio frequency requirements of MIL-STD-461 for class II B equipment.

*3.9 Dimensions. Maximum dimensions including all protrusions of the dummy loads shall not exceed the following:

Type I - 35 inches wide by 45 inches high by 85 inches long

Type II - 40 inches wide by 45 inches high by 120 inches long

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3.10 Weight. Weight of the dummy loads shall not exceed the following:

Type I - 1,800 pounds

Type II - 2,800 pounds

3.11 Color and finish. Unless required to be otherwise finished by MIL-E-16400 (see 3.2), all metal surfaces of the dummy load shall be finished with semigloss enamel conforming to type II of TT-E-485; color shall match Marine Corps green No. 23 (see 6.3). Pretreatment of all surfaces finished in this manner shall be in accordance with MIL-T-704.

3.12 Marking and identification.

3.12.1 Instruction plates. Instruction plates shall be in accordance with MIL-E-16400.

3.12.2 Operating instruction plate. An instruction plate containing operating instructions for the dummy load shall be provided and installed on the inside of the control and meter panel door.

3.12.3 Schematic diagram. A complete schematic diagram shall be provided and installed on the inside of one of the covers or doors (see 3.3.2.1).

3.12.4 Safety plates. Safety and warning plates shall be provided as required to meet requirements of 3.3.1.

3.12.5 Identification plate. An identification plate in accordance with MIL-E-16400 shall be provided and installed on the exterior of the control and meter panel door.

3.13 Government loaned property. Within 90 days after receipt of written request by the contractor to the contracting officer, the Government will furnish one M353 trailer for each type dummy load being supplied under the contract or order. These trailer(s) shall be used by the contractor to determine design requirements of 3.3.5 and for test purposes as specified herein.

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3.14 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The dummy load inspection requirements specified herein are classified as follows:

(a) First Article Inspection (see 4.2.1).

(b) Acceptance Inspection (see 4.2.2).

(c) Sample Inspection (see 4.2.3).

*4.2.1 First article inspection. First article inspection shall consist of all inspection and tests listed under "First Article" in Table I. Inspection and test shall be in the sequence listed in Table I.

*4.2.2 Acceptance inspection. Acceptance inspections and tests specified in Table I shall be performed on each production dummy load submitted for acceptance under the contract.

*4.2.3 Sample inspection. The sample inspection shall consist of the inspections and tests specified in Table I. One dummy load of each type shall be selected at random by the cognizant Government Inspector from each five units of each type dummy load produced. The dummy loads shall have passed the acceptance inspection and tests prior to selection. Failure of the dummy load to pass any inspection or test shall be cause for rejection of the lot. A failure of a dummy load shall be handled as follows.

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TABLE I INSPECTION

Test/Inspection	Test Paragraph	Requirement Paragraph	Acceptance	Sample	First Article
Visual Inspection	4.4.1	3.2,3.3,3.4,3.9, 3.11,3.12,3.14	X		X
Functional	4.4.2	3.2, 3.3	X		X
Electrical	4.4.3	3.2,3.3.7,3.3.8, 3.3.9,3.3.10,3.3.11, 3.3.12,3.3.13,3.3.14, 3.3.15,3.3.16,3.3.17		X	X
Endurance Run	4.4.4	3.2, 3.14		X	X
Towing & Lifting Eye	4.4.5	3.3.4, 3.4			X
Shock	4.4.6	3.4, 3.7			X
Towing	4.4.7	3.3.2.4, 3.4			X
Trailer Transport	4.4.8	3.4, 3.6			X
Rail Transport	4.4.9	3.3.4, 3.4, 3.6			X
High Temperature	4.4.10	3.7			X
Low Temperature	4.4.11	3.7			X
Humidity	4.4.12	3.7			X
Salt Fog	4.4.13	3.7			X
Rain	4.4.14	3.7			X
Enclosure	4.4.15	3.3.2, 3.7			X
Radio Interference	4.4.16	3.8			X
Weight	4.4.17	3.10			X

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4.2.3.1 Random type failure. If the cause of the failure is determined to be of the random type, the failed component shall be replaced and the test program continued.

4.2.3.2 Other failures. If the cause of failure is determined to be other than random type, corrective action shall be taken on test dummy load and on each dummy load in the lot represented by the test unit. At the discretion of the cognizant Government inspector, one of the following programs shall be followed:

(a) The test which was being performed when the failure occurred shall be repeated on the corrected dummy load and the remainder of the test program completed.

(b) The test which was being performed when the failure occurred and any previous tests performed which may have contributed to the failure shall be repeated on the corrected dummy load and the remainder of the test program completed.

(c) The sample inspection shall be repeated on another dummy load which has received the corrective action.

4.3 Test procedures. Procedures for conducting tests shall be prepared by the contractor and submitted for review prior to initiation of any official testing. Testing shall not be performed for record until approval of the test procedure is obtained from the procuring activity.

4.4 Inspection and test. Unless otherwise specified herein, all tests shall be conducted at approximately 20°C and any relative humidity.

4.4.1 Visual inspection. The dummy load shall be inspected to determine conformance to the following requirements:

(a) Materials and parts (see 3.2)

(b) Design (see 3.3)

(c) Construction (see 3.4)

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(d) Maintainability (see 3.4)

(e) Dimensions (see 3.9)

(f) Color and finish (see 3.11)

(g) Identification plate (see 3.12)

(h) Workmanship (see 3.14)

4.4.2 Functional test. The dummy load shall be tested to determine proper performance for each load value from no load to full load at each voltage and frequency input. The specific procedures shall be approved by the procuring activity prior to use. The procedures shall include the 25 percent intermittent overvoltage requirement of 3.3.

4.4.3 Electrical tests. Where external measuring instruments are required, such instruments shall provide an accuracy of reading which is four times better than the accuracy required of the value being measured. The instruments used shall be currently calibrated.

4.4.3.1 Meter calibration. External calibration standards shall be connected to the load bank through the external metering jacks provided on the metering panel. Appropriate inputs shall be applied to the dummy load for 60-Hz and 400-Hz operation for both 120/208- and 240/416-V inputs. The dummy load meters shall be checked against the calibration standards at cardinal points with a minimum of three points on each meter on each range. Meters shall be checked with both 60-Hz and 400-Hz input frequency except the frequency meters shall be checked at their rated input. The meters shall meet the accuracy specified in 3.3.8.

4.4.3.2 Insulation test. A test voltage of 1,500 alternating current volts (Vac) root mean square (rms) at a frequency of 60-Hz shall be applied between the load terminals of the dummy load and equipment ground prior to environmental endurance run testing. Test voltage shall be reduced to 750 Vac rms after environmental or endurance run testing. Any evidence of insulation breakdown shall be cause for rejection.

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4.4.3.3 Overvoltage protection test. The dummy load shall be connected for 120/208-V operation and an input of 240/416 V shall be applied to the input. No damage to the dummy load shall occur as a result of this test and protective devices shall function properly (see 3.3.8 and 3.3.16).

4.4.3.4 Air pressure test. The cooling air path shall be obstructed and the dummy load shall be operated at full load. The air pressure sensing device shall operate and automatically disconnect the dummy load input power (see 3.3.12). No damage to the dummy load shall occur. The cooling fan shall then be disabled and the dummy load operated at full load. The air pressure sensing device shall operate and automatically disconnect the dummy load input power (see 3.3.12). No damage to the dummy load shall occur.

*4.4.3.5 Frequency protection test. The dummy load shall be prepared for operation with an input of 400 Hz, 10 kVA and 0.8 pf. The dummy load shall be energized with a 10-kVA, 60-Hz power source. The protective circuitry specified in 3.3.17 shall operate and prevent damage to the dummy load. Repeat the test with the dummy load prepared for 60-Hz operation and a 400-Hz input. During the above tests the cooling fan motor shall be operated from the dummy load input power and shall not be damaged as a result of improper input frequency.

4.4.4 Endurance run. The dummy load shall be operated at full rated load for 48 hours in accordance with the following voltage, frequency and time schedules. The dummy load shall be set to provide a 0.8 pf. Failure of the dummy load to operate properly during this test shall be cause for rejection.

TIME (hours)	VOLTAGE (Vac)	FREQUENCY (Hz)
12	120/208	60
12	240/416	60
12	120/208	400
12	240/416	400

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4.4.5 Towing and lifting eye test. A 7,000-pound tensile load shall be applied between the towing eyes at various angles to determine conformance to 3.3.2.4. The test shall be repeated on the lifting eyes to determine conformance to 3.3.4. No damage shall occur to the dummy load as a result of this test.

4.4.6 Shock tests. The dummy load shall be subjected to the following drop tests. Minor deformation to the enclosure and skids is allowable; however, such deformation shall not impair proper operation and performance of the dummy load including proper operation of controls, doors, and covers. No structural damage shall be allowed. Upon completion of tests, the dummy load shall meet the requirements of 4.4.2 and 4.4.3.1.

4.4.6.1 Flat drop. The dummy load shall be lifted to a height of 18 inches and allowed to fall freely onto a solid concrete surface.

4.4.6.2 Rotational drop. One edge of the dummy load shall be placed on a 4-inch high block. The opposite edge shall be raised to a height of 18 inches and then allowed to fall freely onto a solid concrete surface. This test shall be performed on each end and each side of the dummy load.

4.4.7 Towing test. The dummy load shall be towed on the skids for a minimum of 1,400 feet in each direction (forward and backward) at a speed of 5 miles per hour (mi/h) over rough terrain. Four right angle turns shall be made on soft dirt. Each turn shall be made from an initial position in which the direction of travel of the towing vehicle is perpendicular to the longitudinal axis of the skids and the towing eye of the vehicle is in line with the leading edge of the dummy load. The dummy load shall show no signs of structural deformation or operational malfunction. Minor dents and superficial surface abrasion is permissible.

4.4.8 Trailer transport test. The dummy load shall be mounted on the M-353 trailer (see 3.13) and transported 500 miles over the Perryman Cross Country Road Course at U. S. Army Proving Grounds, Aberdeen, Maryland. Speeds during this test shall be up to 20 mi/h and there shall be at least ten abrupt stops from speeds of 15 mi/h. No damage shall occur and upon completion of test, the dummy load shall meet the requirements of 4.4.2 and 4.4.3.1.

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4.4.9 Rail transport test. The dummy load shall be installed on a railway flat car and impacted into a stationary mass of five times its weight at a minimum speed of 9 mi/h for a total of four impacts without damage. The dummy load shall be mounted perpendicular to the longitudinal axis of the flat car during two of the impacts and in line with the axis during the other two impacts. The dummy load shall not be packaged or crated during this test, but shall be tied down to the flat car using the lifting and tiedown eyes (see 3.3.4). No damage shall occur as a result of this test and the dummy load shall meet the requirements of 4.4.2 and 4.4.3.1 upon completion of test.

*4.4.10 High temperature test. The dummy load shall be subjected to temperature test in accordance with MIL-STD-810, method 501, procedure II. The dummy load shall be operated at full rated load at 0.8 pf during step 3. The chamber temperature during steps 3 and 5 shall be 55°C in lieu of 49°C. The temperature during step 4 shall be 71°C in lieu of 68°C. The highest operating temperature shall be 55°C. Pretest data shall be the results of the tests specified in 4.4.2 and 4.4.3.1. In addition, the cold resistance of each phase of the resistive load shall be measured with the dummy load connected for 120/208-V operation. The resistance shall be the last measurement made during step 8 and shall be made immediately after removal of power and the resistance shall not vary more than 3 percent from that measured prior to test. During step 3 each load step switch shall be operated to insure proper performance.

*4.4.11 Low temperature test. The dummy load shall be subjected to low temperature test in accordance with MIL-STD-810, method 502. Pretest data shall be the results of the tests specified in 4.4.2 and 4.4.3.1. Operating temperature shall be -40°C. Storage temperature shall be -60°C.

*4.4.12 Humidity test. The dummy load shall be subjected to humidity test in accordance with MIL-STD-810, method 507, procedure I. Pretest data shall be the results of the tests specified in 4.4.2 and 4.4.3.1. These tests shall be repeated at the end of each 24-hour cycle.

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*4.4.13 Salt fog test. The dummy load shall be subjected to salt fog test in accordance with MIL-STD-810, method 509, procedure I. Pretest data shall be the results of the tests specified in 4.4.2 and 4.4.3.1. Upon completion of test, the dummy load shall be operated at full load at 120/208 Vac, 60 Hz and 0.8 pf for a period of 4 hours.

*4.4.14 Rain test. The dummy load shall be subjected to the rain test in accordance with MIL-STD-810, method 506, procedure I. Pretest data shall be the results of the tests specified in 4.4.2 and 4.4.3.1. The dummy load shall be operated at full load with a 0.8 pf during the test.

*4.4.15 Enclosure test. The dummy load shall be subjected to the enclosure test specified in MIL-E-16400 for dripproof equipment. Leakage within the enclosure which would impair proper operation shall be cause for failure to pass the test.

*4.4.16 Radio interference suppression test. The dummy load shall be subjected to radiated emission test in accordance with method RE 02 of MIL-STD-462.

*4.4.17 Weight test. The dummy load shall be weighed on a calibrated scale and the weight shall not exceed limits specified in 3.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packing shall be level A or C as specified (see 6.2).

5.1.1 Level A. The dummy load shall be preserved and packaged in accordance with level A requirement of MIL-P-16298.

5.1.2 Level C. Complete dummy load shall be preserved and packaged to afford protection against deterioration and damage.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. The complete dummy load preserved and packaged as specified in 5.1.1 requires no additional packing.

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5.2.2 Rigidly supported units. Rigidly supported units shall not be

5.2.2 Level C. The complete dummy loads shall be packed to insure safe arrival at destination at lowest rates in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The dummy loads covered by this specification are intended for use in simulating electrical loads for engine generator sets at field maintenance organizations

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification

(b) Type required (see 1.2)

(c) Whether first article is required (see 3.1)

(d) Level of preservation, packaging and packing required (see 5.1 and 5.2)

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions C-01, C-03, C-04, E-04, E-11, E-12, E-13, H-01, H-06, H-07, R-02, R-03, R-04, T-01, T-02, T-03, T-04, V-01, V-08, V-09, V-11 and V-16 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from Commandant of the Marine Corps, Headquarters U. S. Marine Corps (Code CSY), Washington, D.C., 20380.

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6.4 Road test. For information on the Perryman Road Course Tests, contact the Director, Development and Proof Services, Aberdeen Proving Grounds, Maryland.

6.5 First article. First article testing shall be witnessed by a representative of the procuring activity. Approval of the first article shall be by the contracting officer.

*6.6 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - MC
Project No. 6115-N324

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OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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